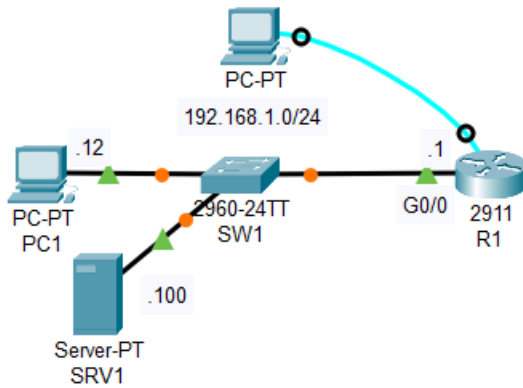




Topology

R1 username: jeremy, PW: ccna, enable PW: ccna

1. Connect to R1's console port using PC2:

- Shut down the G0/0 interface

```

PC2
Physical Config Desktop Programming Attributes
Terminal
If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

User Access Verification

Username: jeremy
Password:

R1>en
Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int g0/0
R1(config-if)#shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
  
```

```
R1(config-if)#do sh ip int br
Interface          IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0 192.168.1.1     YES manual administratively down down
GigabitEthernet0/1 unassigned      YES unset  administratively down down
GigabitEthernet0/2 unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
```

There are two Syslog messages, one to convey the status change of the interface to 'administratively down' and another for the protocol 'down'.

- After you receive a syslog message, re-enable the interface.

```
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

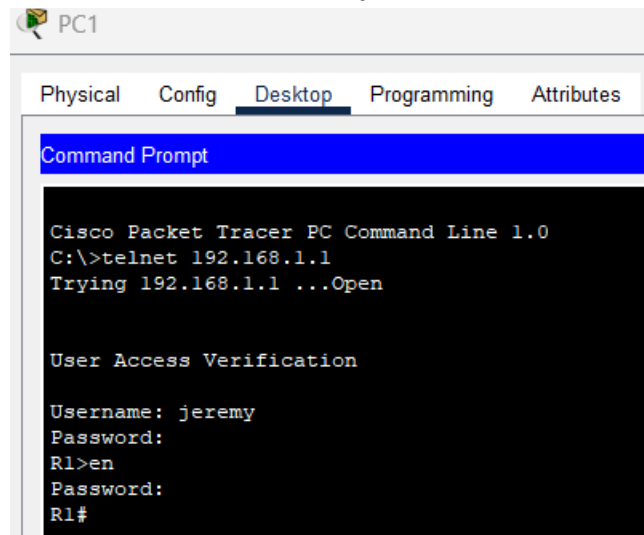
- What is the severity level of the syslog messages?

5 (Notice/Notification)

- Enable timestamps for logging messages

```
R1(config)#service timestamps log datetime msec
```

2. Telnet from PC1 to R1's G0/0 interface



- Enable the unused G0/1 interface

```
R1(config)#int g0/1
R1(config-if)#no shut

R1(config-if)#
```

- Why does no syslog message appear?

Syslog is not enabled when connecting via the VTY lines.

- Enable logging to the VTY lines for the current session.

```

R1(config-if)#do terminal monitor
R1(config-if)#shut

R1(config-if)#
*Feb 28, 23:12:49.1212: %LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to
administratively down

```

*there is no 'logging monitor' command in packet tracer, but it's enabled by default

3. Enable logging to the buffer, and configure the buffer size to 8192 bytes.

```

R1(config-if)#do sh logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 9 messages logged, xml disabled,
filtering disabled
Monitor logging: level debugging, 1 messages logged, xml disabled,
filtering disabled
Buffer logging: disabled, xml disabled,
filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

```

Logging buffer is disabled by default.

```

R1(config)#logging buffered 8192
R1(config)#do sh logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 9 messages logged, xml disabled,
filtering disabled
Monitor logging: level debugging, 1 messages logged, xml disabled,
filtering disabled
Buffer logging: level debugging, 0 messages logged, xml disabled,
filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

```

The default level is debugging, so all messages will be saved to the buffer. The trap logging: level informational means that if a syslog server is configured, messages of informational severity or higher will be sent.

4. Enable logging to the syslog server SRV1 with a level of 'debugging'.

```

R1(config)#logging host 192.168.1.100
R1(config)#logging trap debugging

```

```

R1(config)#int g0/1
R1(config-if)#no shut

R1(config-if)#
*Feb 28, 23:19:13.1919: %LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
R1(config-if)#shut

R1(config-if)#
*Feb 28, 23:19:18.1919: %LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to
administratively down

```

SRV1

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

Syslog

Syslog

Service

On

Off

	Time	HostName	Message
1	02.28.1993 11:19:13.656 PM	192.168.1.1	%LINK-5-CHANGED: Interface ...
2	02.28.1993 11:19:18.484 PM	192.168.1.1	%LINK-5-CHANGED: Interface ...
3	02.28.1993 11:19:45.678 PM	192.168.1.1	%SYS-5-CONFIG_I: Configured from console by console